



TRIOS CYBER

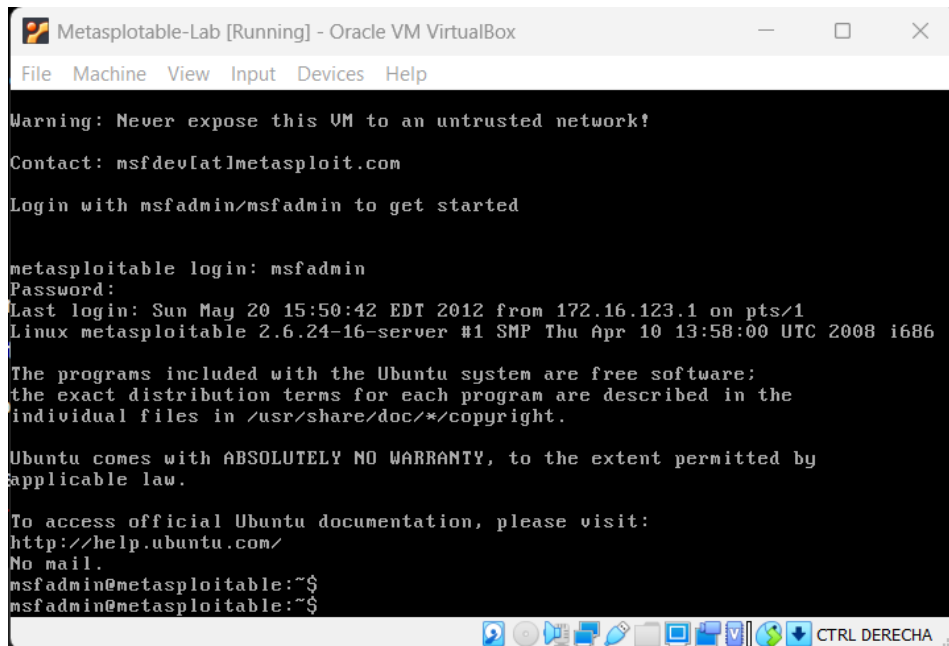
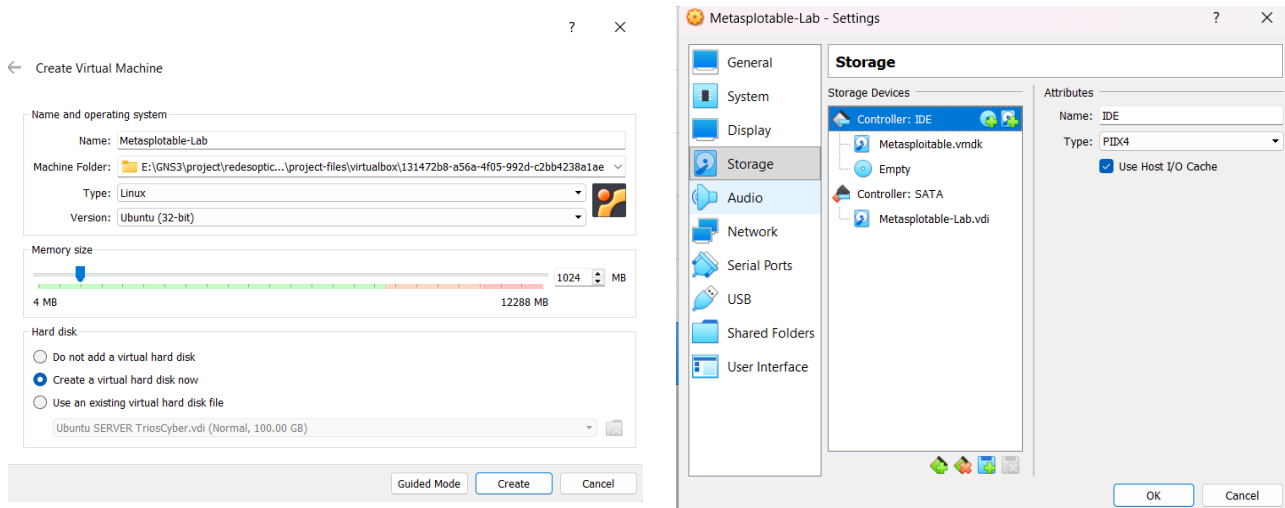
LABORATORY PRACTICE 2

Nmap Scanning and Service Discovery

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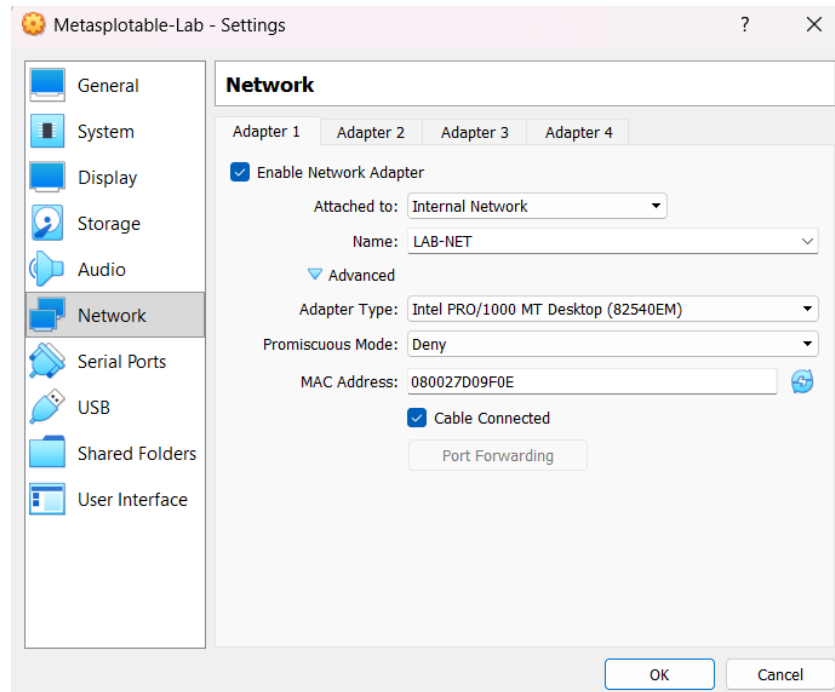
1. Metasploitable 2 Virtual Machine Setup

A Metasploitable 2 virtual machine was created in Oracle VirtualBox and the existing Metasploitable virtual disk was attached to the machine. The target system was started successfully and access to the default laboratory account was verified.



2. Isolated Network Configuration

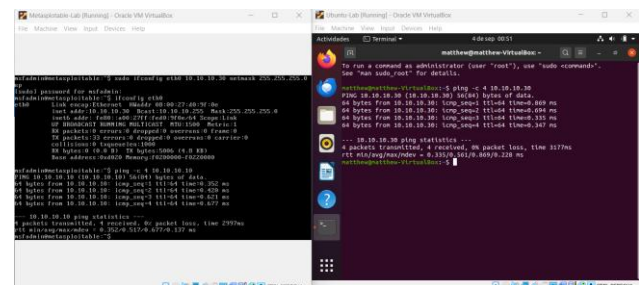
The Metasploitable target was connected to the same internal VirtualBox network named LAB-NET as the Ubuntu scanning machine. This configuration kept the activity inside the isolated laboratory environment.



3. Target IP Configuration and Connectivity

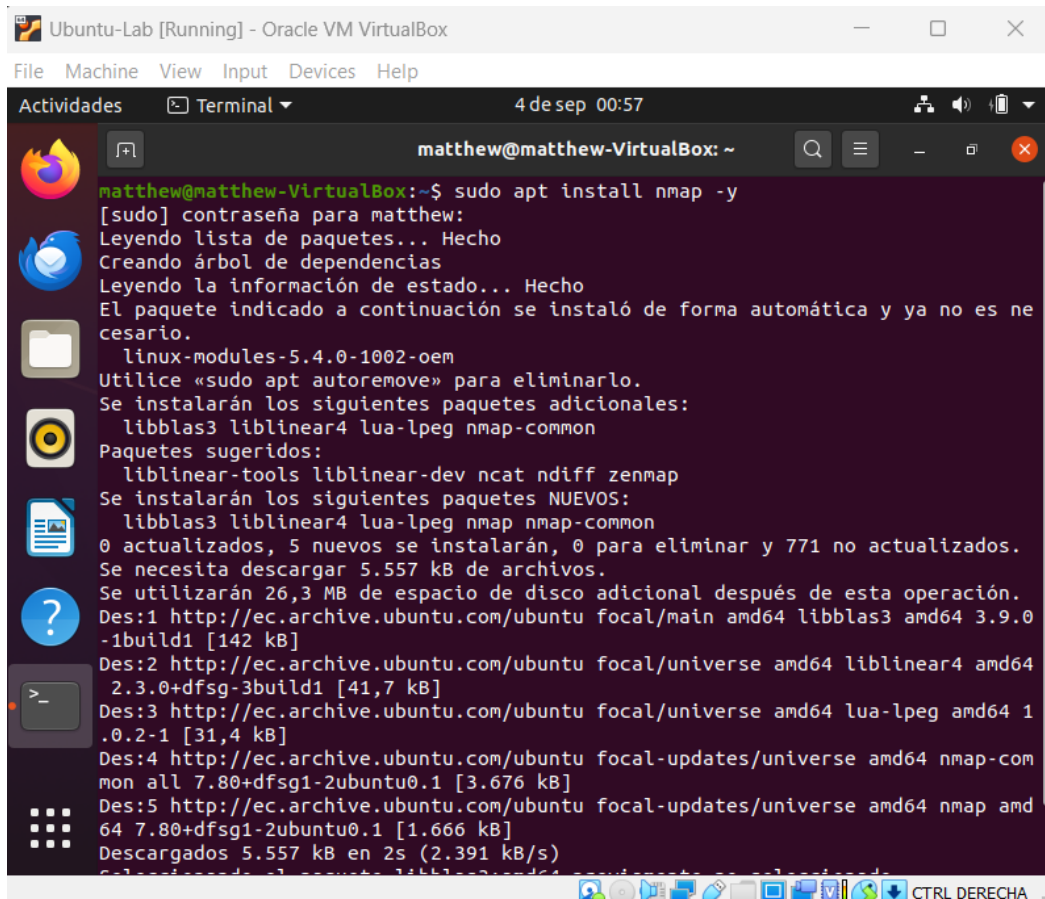
A static IPv4 address of 10.10.10.30/24 was assigned to the Metasploitable interface. Connectivity with the Ubuntu scanning machine at 10.10.10.10 was then verified using ping before the Nmap scans were performed.

```
Metasploitable-Lab [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Z: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:d0:9f:0e brd ff:ff:ff:ff:ff:ff
    inet6 fe80::a00:27ff:fe40:9f0e/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$ s * Reloading OpenBSD Secure Shell server's configuration
..done.
* Reloading Postfix configuration...
..done.
msfadmin@metasploitable:~$ sudo ifconfig eth0 10.10.10.30 netmask 255.255.255.0 up
[sudo] password for msfadmin:
msfadmin@metasploitable:~$ ifconfig eth0
eth0
    Link encap:Ethernet  HWaddr 08:00:27:d0:9f:0e
    inet addr:10.10.10.30  Bcast:10.10.10.255  Mask:255.255.255.0
    inet6 addr: fe80::a00:27ff:fe40:9f0e/64 Scope:Link
    UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
    RX packets:0 errors:0 dropped:0 overruns:0 frame:0
    TX packets:33 errors:0 dropped:0 overruns:0 carrier:0
    collisions:0 txqueuelen:1000
    RX bytes:0 (0.0 B)  TX bytes:5006 (4.8 KB)
    Base address:0xd020 Memory:f0200000-f0220000
msfadmin@metasploitable:~$
```



4. Nmap Installation

Nmap was installed on the Ubuntu scanning machine using the APT package manager. The installation provided the network scanning utility required for host discovery, port scanning, and service identification.



```
matthew@matthew-VirtualBox: ~$ sudo apt install nmap -y
[sudo] contraseña para matthew:
Leyendo lista de paquetes... Hecho
Creando árbol de dependencias
Leyendo la información de estado... Hecho
El paquete indicado a continuación se instaló de forma automática y ya no es necesario.
  linux-modules-5.4.0-1002-oem
Utilice «sudo apt autoremove» para eliminarlo.
Se instalarán los siguientes paquetes adicionales:
  libblas3 liblinear4 lua-lpeg nmap-common
Paquetes sugeridos:
  liblinear-tools liblinear-dev ncat ndiff zenmap
Se instalarán los siguientes paquetes NUEVOS:
  libblas3 liblinear4 lua-lpeg nmap nmap-common
0 actualizados, 5 nuevos se instalarán, 0 para eliminar y 771 no actualizados.
Se necesita descargar 5.557 kB de archivos.
Se utilizarán 26,3 MB de espacio de disco adicional después de esta operación.
Des:1 http://ec.archive.ubuntu.com/ubuntu focal/main amd64 libblas3 amd64 3.9.0-1build1 [142 kB]
Des:2 http://ec.archive.ubuntu.com/ubuntu focal/universe amd64 liblinear4 amd64 2.3.0+dfsg-3build1 [41,7 kB]
Des:3 http://ec.archive.ubuntu.com/ubuntu focal/universe amd64 lua-lpeg amd64 1.0.2-1 [31,4 kB]
Des:4 http://ec.archive.ubuntu.com/ubuntu focal-updates/universe amd64 nmap-common all 7.80+dfsg1-2ubuntu0.1 [3.676 kB]
Des:5 http://ec.archive.ubuntu.com/ubuntu focal-updates/universe amd64 nmap amd64 7.80+dfsg1-2ubuntu0.1 [1.666 kB]
Descargados 5.557 kB en 2s (2.391 kB/s)
Se instalarán los siguientes paquetes adicionales:
  libblas3 liblinear4 lua-lpeg nmap-common
```

5. Live Host Discovery

A host discovery scan was performed on the isolated 10.10.10.0/24 network. The scan identified two live systems: the Ubuntu scanner at 10.10.10.10 and the Metasploitable target at 10.10.10.30.

```
matthew@matthew-VirtualBox:~$ nmap -sn 10.10.10.0/24
Starting Nmap 7.80 ( https://nmap.org ) at 2026-09-04 00:58 -05
Nmap scan report for matthew-VirtualBox (10.10.10.10)
Host is up (0.000046s latency).
Nmap scan report for 10.10.10.30
Host is up (0.00081s latency).
Nmap done: 256 IP addresses (2 hosts up) scanned in 3.20 seconds
matthew@matthew-VirtualBox:~$
```

6. Open Port and Service Detection

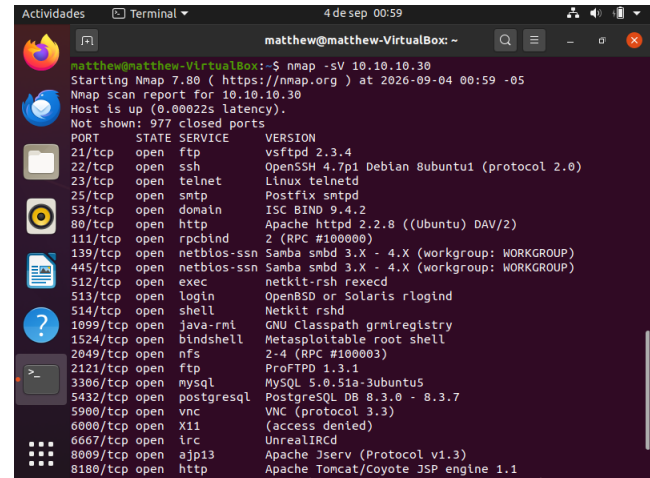
Nmap was run only against the Metasploitable target at 10.10.10.30. Multiple open TCP ports were identified, and service version detection was used to determine the applications listening on those ports.

```

21/tcp open  ftp
22/tcp open  ssh
23/tcp open  telnet
25/tcp open  smtp
53/tcp open  domain
80/tcp open  http
111/tcp open rpcbind
139/tcp open netbios-ssn
445/tcp open microsoft-ds
512/tcp open exec
513/tcp open login
514/tcp open shell
1524/tcp open ingreslock
2049/tcp open nfs
2121/tcp open ceph-proxy-ftp
3306/tcp open mysql
3632/tcp open distccd
5432/tcp open postgres
5900/tcp open vnc
6000/tcp open x11
6667/tcp open irc
8009/tcp open ajp13

Nmap done: 1 IP address (1 host up) scanned in 13.104 seconds
msfadmin@metasploitable:~$

```



```

matthew@matthew-VirtualBox:~$ nmap -sV 10.10.10.30
Starting Nmap 7.80 ( https://nmap.org ) at 2026-09-04 00:59 -05
Nmap scan report for 10.10.10.30
Host is up (0.00022s latency).
Not shown: 977 closed ports
PORT      STATE SERVICE        VERSION
21/tcp    open  ftp            vsftpd 2.3.4
22/tcp    open  ssh            OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet         Linux telnetd
25/tcp    open  smtp           Postfix smtpd
53/tcp    open  domain         ISC BIND 9.4.2
80/tcp    open  http           Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind        2 (RPC #100000)
139/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn    Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec           netkit-rsh rexecd
513/tcp   open  login          OpenBSD or Solaris rlogind
514/tcp   open  shell          Netkit rshd
1099/tcp  open  java-rmi       GNU Classpath grmiregistry
1524/tcp  open  bindshell      Metasploitable root shell
2049/tcp  open  nfs            2-4 (RPC #100003)
2121/tcp  open  ftp            ProFTPD 1.3.1
3306/tcp  open  mysql          MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql     PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc            VNC (protocol 3.3)
6000/tcp  open  x11            (access denied)
6667/tcp  open  irc            UnrealIRCd
8009/tcp  open  ajp13          Apache Jserv (Protocol v1.3)
8180/tcp  open  http           Apache Tomcat/Coyote JSP engine 1.1

```

```

8180/tcp open  http           Apache Tomcat/Coyote JSP engine 1.1
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs:
Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 11.48 seconds

```

7. Service Exposure Review

The discovered services were consistent with the intentionally vulnerable design of Metasploitable 2. Services such as FTP, SSH, Telnet, HTTP, SMB, NFS, MySQL, PostgreSQL, VNC, IRC, AJP13, and Apache Tomcat were exposed. No additional unexpected live host was identified; however, legacy remote-access services and externally reachable database services would be considered unnecessary exposure in a normal production environment.

8. Conclusions

The Nmap activity was successfully completed inside the isolated laboratory environment. The Metasploitable target was identified as a live host, multiple open ports and network services were detected, and the exposed services were reviewed against the intentionally vulnerable configuration. The results demonstrated how service enumeration can be used to identify unnecessary or potentially risky network exposure without scanning systems outside the authorized lab.